3. EQUILIBRIUM

I follow the bulk of the search literature by letting $\beta \in [0,1)$ and assuming the instantaneous wage $w(x)$ and labour supply n solve max $[W(x)-U]^{\beta}[J(x)-V]^{1-\beta}$ at all times. The optimal choice of hours is

$$n(x) = \begin{cases} k & \text{if } \phi < x \\ 0 & \text{if } x \leq \phi, \end{cases} \tag{8}$$

and substituting it into the flow profit function gives $\pi(x) = (x-\phi-c)k - w(x)$ for all x. The first-order condition for the instantaneous wage, $w(x)$, is

$$(1-\beta)[W(x)-U] = \beta[J(x)-V]. \tag{9}$$

Letting $S(x) = J(x) + W(x) - U - V$ denote the surplus from a match, notice that (9) implies $J(x) = (1-\beta)S(x)$ and $W(x) - U = \beta S(x)$. These, together with (4), (5), and (7), imply

$$(r+\delta+\lambda)S(x) = (x-\phi-c)k - rU + \lambda \int \max[S(z),0]dG(z),$$

where

$$rU = b + \frac{\beta}{1-\beta}kc\theta. \tag{10}$$

In deriving (10) I have already acknowledged that free entry of firms will cause $rV = 0$ in equilibrium. Since $S'(x) = \frac{k}{r+\delta+\lambda} > 0$, there exists a unique R such that $S(x) > 0$ iff $x > R$. Hence, a firm–worker pair destroys an existing match (and chooses not to form a new match) if it draws a productivity $x < R$. With this reservation strategy, the surplus can be written as

$$(r+\delta+\lambda)S(x) = (x-\phi-c)k - rU + \lambda \int_R S(z)dG(z). \tag{11}$$

The value functions and (9) can be combined to obtain the instantaneous wages and profit:

$$w(x) = \beta(x-\phi-c)k + (1-\beta)rU \tag{12}$$

$$\pi(x) = (1-\beta)[(x-\phi-c)k - rU]. \tag{13}$$

Intuitively, the wage is a weighted average of output (net of the rental on capital and the variable and fixed costs) and the worker's reservation wage.

Next, I characterize the job-creation and job-destruction decisions as summarized by θ and R, respectively. Evaluating (11) at $x = R$ yields $\lambda \int_R S(z)dG(z) = rU - (R-\phi-c)k$. Notice that since the expected capital gain on the L.H.S. is positive, at $x = R$ net output is smaller than the worker's reservation wage. Thus, (12) and (13) imply that $w(R) < rU$ and $\pi(R) < 0$: workers and firms sometimes tolerate instantaneous pay-offs below those they could get by separating, in anticipation of future productivity improvements.[16] Substituting this simpler expression for the expected capital gain term into (11) gives

$$S(x) = \frac{x-R}{r+\delta+\lambda}k. \tag{14}$$

Appendix B I redo the whole analysis with a more general specification for the fixed cost and show that $\frac{\partial C(x,\phi)}{\partial x} < 0$ is all that is needed. But again, see Section 8 for alternative specifications that do not rely on firms having to bear *any* fixed cost C.

16. This feature of the model is a consequence of the costly and time-consuming meeting process, as noted by Mortensen and Pissarides (1994).

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Evaluating (11) at $x = R$ and using (14) to substitute $S(\cdot)$ yields what is usually referred to as the job-destruction condition:

$$R - \phi - c - \left(\frac{b}{k} + \frac{\beta}{1-\beta}c\theta\right) + \frac{\lambda}{r+\delta+\lambda}\int_R (x-R)dG(x) = 0. \tag{15}$$

As is standard, the destruction decision is independent of scale if b is. The natural interpretation of b is that it is unemployment insurance income. Along these lines, if one lets $b = \tau_b E_G[w(x) \mid x \geq R]$, where $\tau_b \in [0,1)$ is the replacement rate, then $b = \hat{b}k$, with

$$\hat{b} = \frac{\tau_b \beta[\tilde{x}(R) - \phi - c + c\theta]}{1-(1-\beta)\tau_b},$$

and $\tilde{x}(R) \equiv E_G[x \mid x \geq R] = [1-G(R)]^{-1}\int_R x dG(x)$. Under this specification, b is linear in k so (15) is independent of k and becomes

$$R - \frac{\tau_b \beta \tilde{x}(R)}{1-(1-\beta)\tau_b} - \frac{(1-\tau_b)(\phi+c)}{1-(1-\beta)\tau_b} - \frac{\beta c\theta}{(1-\beta)[1-(1-\beta)\tau_b]} + \frac{\lambda}{r+\delta+\lambda}\int_R (x-R)dG(x) = 0.$$

In what follows I will always abstract from scale effects caused by unemployment income b by assuming it is a fraction of the average going wage.

Substituting the equilibrium condition $rV = 0$ in (6) implies $(1-\beta)\int_R S(x)dG(x) = \frac{ck}{q(\theta)}$. That is, the expected profit from a filled job equals the expected hiring cost in an equilibrium with free entry. Using (14) to substitute $S(\cdot)$ out of this expression yields what is often referred to as the job-creation condition:

$$\frac{1-\beta}{r+\delta+\lambda}\int_R (x-R)dG(x) - \frac{c}{q(\theta)} = 0. \tag{16}$$

The job-creation and job-destruction conditions jointly determine R and θ, and under the maintained assumptions, they are independent of scale, k.[17] For given c and ϕ, an equilibrium is a vector $[\theta, R, H, U, w, u, K]$ such that (θ, R) jointly solve (15) and (16); given (θ, R), H satisfies (3); U is given by (10); w is given by (12); and u is given by (2). In addition, the market for capital services should clear, so the aggregate supply of capital K must satisfy $K = [1-(1-\theta)u]k$, where the R.H.S. is the total demand for capital (coming from both matched and unmatched firms). Note that in parametrizations that result in $R < \phi$, the capital and workers in matches with realizations in $[R, \phi)$ remain employed but are not engaged in production. The firms in these states have excess capacity and hoard labour. The following section provides a sharper characterization of aggregate outcomes for a particular distribution of idiosyncratic shocks.

4. AGGREGATION

Let K_e denote the capital in place at all the firms with filled jobs, that is,

$$K_e = \frac{1-u}{1-(1-\theta)u}K. \tag{17}$$

17. See Lemma 1 in Appendix B for conditions under which the pair (θ, R) that solves (15) and (16) exists and is unique.

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